

Logic Building Assessment Day 4

Ques1

```
import java.util.Scanner;
```

```
public class PrintNumbersToN {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("enter a number:");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            System.out.print(i+" ");
```

```
        }
```

```
    }
```

```
}
```

Ques2

```
import java.util.Scanner;
```

```
public class PrintMultipleOfThree {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.println("enter a number:");  
        int n = sc.nextInt();  
        for (int i = 3; i <= n; i++) {  
            System.out.print(i+" ");  
            i=i+3;  
        }  
    }  
}
```

Ques3

```
import java.util.Scanner;

public class PrintMultipleOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("enter a number:");
        int n = sc.nextInt();
        for (int i = 3; i <= n; i++) {
            System.out.print(i+" ");
            i=i+3;
        }
    }
}
```

Ques4

```
import java.util.Scanner;

public class PrintEvenNumbers {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("enter a number:");
        int n = sc.nextInt();
        for (int i = 2; i <= n; i+= 2) {
            System.out.print(i + " ");
        }
    }
}
```

Ques5

```
import java.util.Scanner;
```

```
public class SumOfOdd {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("enter a number:");
```

```
        int n = sc.nextInt();
```

```
        int sum = 0;
```

```
        for (int i = 1; i <= n; i = i + 3) {
```

```
            sum = sum + i;
```

```
        }
```

```
        System.out.println("The sum of odd numbers from 1 to" + n + " is:" + 25);
```

```
    }
```

```
}
```

Ques 6

```
import java.util.Scanner;
```

```
public class PrintAllElements {  
    public static void main(String[] args) {  
        int[] arr = new int[5];  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("enter 5 integers:");  
  
        for (int i = 0; i < arr.length; i++) {  
            arr[i] = sc.nextInt();  
  
        }  
  
        for (int i : arr) {  
            System.out.print(i + " ");  
        }  
  
    }  
  
}
```

Ques 7

```
import java.util.Scanner;
```

```
public class SumOfAll {
```

```
    public static void main(String[] args) {
```

```
        int[] arr = new int[5];
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("enter 5 integers:");
```

```
        for (int i = 0; i < arr.length; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        int sum = 0;
```

```
        for (int i : arr) {
```

```
            sum = sum + i;
```

```
        }
```

```
        System.out.println("The sum of all number is:" + sum);
```

```
    }
```

```
}
```

Ques 8

```
import java.util.Scanner;
```

```
public class PrintAllNames {  
    public static void main(String[] args) {  
        System.out.print("Enter 4 names: ");  
        Scanner sc = new Scanner(System.in);  
        String[] names = new String[4];  
        for (int i = 0; i < 4; i++) {  
            names[i] = sc.next();  
  
        }  
        for (String name : names) {  
            System.out.println(name);  
  
        }  
    }  
}
```

Ques 9

```
import java.util.Arrays;
```

```
public class FindLargestElement {
```

```
    public static void main(String[] args) {
```

```
        int []arr={89,12,45,67,23};
```

```
        Arrays.sort(arr);
```

```
        System.out.println("The largest element is:"+arr[4]);
```

```
    }
```

```
}
```


Ques 10

```
import java.util.Scanner;
```

```
public class FindAverage {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.println("Enter array size");  
        int size = sc.nextInt();  
        int[] arr = new int[size];  
        System.out.println("enter" + size + " integers:");  
        for (int i = 0; i < arr.length; i++) {  
            arr[i] = sc.nextInt();  
  
        }  
        int sum = 0;  
        for (int i : arr) {  
            sum = sum + i;  
        }  
        System.out.println("The average of the numbers is:" + (((float) sum) / size));  
  
    }  
  
}
```

Ques 11

```
import java.util.Scanner;
```

```
public class PositiveNegative {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int[] arr = new int[6];
```

```
        int positive = 0;
```

```
        int negative = 0;
```

```
        System.out.println("Enter 6 integers:");
```

```
        for (int i = 0; i < 6; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
            if (arr[i] > 0) {
```

```
                positive++;
```

```
            } else if (arr[i] < 0) {
```

```
                negative++;
```

```
            }
```

```
        }
```

```
        System.out.println("Positive numbers: " + positive);
```

```
        System.out.println("Negative numbers: " + negative);}}
```

Ques 12

```
import java.util.Arrays;
```

```
public class SortTheArray {
```

```
    public static void main(String[] args) {
```

```
        int []arr={89,12,45,67,23};
```

```
        Arrays.sort(arr);
```

```
        System.out.println(Arrays.toString(arr));
```

```
    }
```

```
}
```

Ques 13

```
import java.util.Arrays;
```

```
import java.util.List;
```

```
public class FindElement {
```

```
    public static void main(String[] args) {
```

```
        List<Integer>arr=Arrays.asList(1,2,3,4,5);
```

```
        int target=6;
```

```
        if (arr.contains(target)) {
```

```
            System.out.println("Found");
```

```
        } else {
```

```
            System.out.println("Not Found");
```

```
        }
```

```
    }
```

```
}
```

Ques 14

```
import java.util.Arrays;
```

```
import java.util.Scanner;
```

```
public class FindTheIndex {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter array size");
```

```
        int size = sc.nextInt();
```

```
        int[] arr = new int[size];
```

```
        System.out.println("enter " + size + " integers:");
```

```
        for (int i = 0; i < arr.length; i++) {
```

```
            arr[i] = sc.nextInt();
```

```
        }
```

```
        System.out.println("Enter the number to search:");
```

```
        int target = sc.nextInt();
```

```
        System.out.println("the number " + target + " is found at index " + Arrays.binarySearch(arr,  
target));
```

```
    }
```

```
}
```

Ques 24

```
public class ReverseString {  
    public static void main(String[] args) {  
        String str="hello";  
        for (int i =str.length()-1;i>=0;i--) {{  
            System.out.print(str.charAt(i));  
        }  
  
        }  
    }  
}
```

Ques 15

```
import java.util.Scanner;
```

```
public class Pattern15 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print(i);
```

```
                if (j < i) {
```

```
                    System.out.print("*");
```

```
                }
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
        for (int i = n; i >= 2; i--) {
```

```
for (int j = 1; j <= i; j++) {  
    System.out.print(i);  
  
    if (j < i) {  
        System.out.print("*");  
    }  
}  
  
System.out.println();  
}  
}  
}
```


Ques 16

```
import java.util.Scanner;
```

```
public class Pattern16 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print(j);
```

```
                if (j < i) {
```

```
                    System.out.print("*");
```

```
                }
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }}
```

Ques 17

```
import java.util.Scanner;
```

```
public class Pattern17 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print(2 * j - 1);
```

```
                if (j < i) {
```

```
                    System.out.print("*");
```

```
                }
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }}
```

Ques 18

```
package Pattern;
```

```
import java.util.Scanner;
```

```
public class Pattern18 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            for (int j = 1; j <= n; j++) {
```

```
                System.out.print(i);
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

Ques 19

```
package Pattern;
```

```
import java.util.Scanner;
```

```
public class Pattern19 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print(i);
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

Ques 20

```
package Pattern;
```

```
import java.util.Scanner;
```

```
public class Pattern20 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print(j);
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

Ques 21

```
package Pattern;
```

```
import java.util.Scanner;
```

```
public class Pattern21 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
        int n = sc.nextInt();
```

```
        int num = 1;
```

```
        for (int i = 1; i <= n; i++) {
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print(num + " ");
```

```
                num++;
```

```
            }
```

```
        System.out.println();}}
```

Ques 22

```
package Pattern;
```

```
import java.util.Scanner;
```

```
public class Pattern22 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
        int n = sc.nextInt();
```

```
        for (int i = 1; i <= n; i++) {
```

```
            for (int j = 1; j <= n; j++) {
```

```
                if (i == 1 || i == n || j == 1 || j == n) {
```

```
                    System.out.print("*");
```

```
                } else {
```

```
                    System.out.print(" ");
```

```
                }
```

```
            }
```

```
        System.out.println();}}
```

Ques 23

```
package Pattern;
```

```
import java.util.Scanner;
```

```
public class Pattern23 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter odd number: ");
```

```
        int n = sc.nextInt();
```

```
        // Increasing part
```

```
        for (int i = 1; i <= n; i += 2) {
```

```
            for (int j = 1; j <= (n - i) / 2; j++) {
```

```
                System.out.print(" ");
```

```
            }
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print("*");
```

```
            }
```

```
        System.out.println();
```



```
}
```

```
// Decreasing part
```

```
for (int i = n - 2; i >= 1; i -= 2) {
```

```
    for (int j = 1; j <= (n - i) / 2; j++) {
```

```
        System.out.print(" ");
```

```
    }
```

```
    for (int j = 1; j <= i; j++) {
```

```
        System.out.print("*");
```

```
    }
```

```
    System.out.println();
```

```
}
```

```
}
```

```
}
```

Ques 24

```
public class ReverseString {  
    public static void main(String[] args) {  
        String str="hello";  
        for (int i =str.length()-1;i>=0;i--) {{  
            System.out.print(str.charAt(i));  
        }  
  
        }  
    }  
}
```

Ques 25

```
public class CountVowels {  
    public static void main(String[] args) {  
        String str = "programmimg";  
        int count = 0;  
  
        for (int i = str.length() - 1; i >= 0; i--) {  
  
            if (str.charAt(i) == 'a' || str.charAt(i) == 'e' ||  
                str.charAt(i) == 'i' || str.charAt(i) == 'o' || str.charAt(i) == 'u') {  
                count++;  
  
            }  
        }  
        System.out.println("The number of vowels in '" + str + "' is " + count);  
    }  
}
```

Ques 26

```
public class CheckPalindrome {  
    public static void main(String[] args) {  
        String str = "madam";  
        String rev = "";  
        for (int i = str.length() - 1; i >= 0; i--) {  
            rev = rev + str.charAt(i);  
        }  
        if (str.equals(rev)) {  
            System.out.println("The string '" + str + "' is a palindrome");  
        } else {  
            System.out.println("String is not palindrome");  
        }  
    }  
}
```

Ques 27

```
public class StringLiteral {  
    public static void main(String[] args) {  
        String str1 = "hello";  
        String str2 = "hello";  
        if (str1==str2)  
            System.out.println("Both variables points to the same object: " + (str1 == str2));  
    }  
}
```

Ques 28

```
public class StringNewKeyword {  
    public static void main(String[] args) {  
        String s1 = new String("hello");  
        String s2 = new String("hello");  
        System.out.println("Using ==: " + (s1 == s2));  
        System.out.println("Using .equals(): " + s1.equals(s2));  
    }  
  
}
```

Ques 29

```
public class StringConcat {  
    public static void main(String[] args) {  
        String str1 = "hello";  
        String str2 = "world";  
        String str3 = str1 + str2;  
        System.out.println("Is str3 pointing to the same object as str1? " + //  
            ""+(str3==(str1+str2)));  
    }  
  
}
```

Ques 30

```
public class StringIntern {  
    public static void main(String[] args) {  
        String str1 = new String("hello");  
        String str2 = str1.intern();  
        String str3 = "hello";  
        if (str2 == str3) {  
            System.out.println("Is str2 and str3 pointing to the same object?" + (str2 == str3));  
        }  
    }  
}
```


Ques 31

```
public class MultipleStringLiteral {  
    public static void main(String[] args) {  
        String str1 = "java";  
        String str2 = "java";  
        String str3 = "java";  
        if (str1 == str2 && str2 == str3) {  
            System.out.println("All strings point to the same object:" + (str1 == str2 && str2 == str3));  
        }  
        else{  
            System.out.println("all string are not pointing the same obj");  
        }  
    }  
}
```